

Dunnhumby Complete Journey

A targeted-coupon recommender

WHO	Category manager, US grocery retailer
WHAT	Top-3 coupon-campaign ranking per household
WHEN	2-year temporal split
WHERE	US grocery chain
WHY	~70% of untargeted coupons wasted — ranking is money
HOW	RFM baseline → implicit ALS + hybrid

~70%

of untargeted coupon impressions go to households that would not buy the category, or would have bought at full price.

THE DATASET

8 CSVs · ~300 MB · 102 weeks of history

2,500 loyalty households · 92k SKUs · 30 campaigns · coupon send + redemption ledger + causal in-store promo flags

Kaggle mirror · [frtgnn/dunnhumby-the-complete-journey](#) · one download, no signup

Research question + evaluation protocol

RESEARCH QUESTION

*"Under a temporal split of Dunnhumby's panel, does a **weighted-implicit-ALS** model on household × commodity spend improve **NDCG@3** and **Recall@3** over an **RFM-segment popularity baseline**, and how does it trade off **category coverage** and **coupon-redemption uplift**?"*

PROTOCOL = split + candidate set + ground truth + filtering

SPLIT

Temporal

Train w1–91 · Val w92–96 · Test w97–102.

CANDIDATE SET

~30 active campaigns

The small, clean pool of campaigns live in the test window.

GROUND TRUTH

campaign_table.csv

Which campaigns the retailer assigned to each household. Hit = our top-3 contains the assigned campaign.

FILTERING

Households with ≥1 signal

Drop households with zero purchases and zero assigned campaigns in the test window

Baseline vs improved

BASELINE

RFM-segment popularity

Method Segment households by Recency / Frequency / Monetary. Recommend the top-3 campaigns most redeemed inside the segment.

Signal Aggregate segment behaviour only.

Strength Instant, interpretable, robust cold-household fallback.

Weakness Ignores 2 years of granular history; no per-household signal beyond segment.



IMPROVED

Weighted implicit ALS + campaign matching

Method Hu-Koren-Volinsky ALS on log-spend household × commodity (dim=64). Rank campaigns by predicted score of their covered commodities.

Signal Every purchase, spend-weighted, 2 years of history.

Strength Captures latent taste; extends cleanly to a LightFM hybrid with product hierarchy.

Weakness Weak on cold households; less interpretable than RFM.

Next Steps

MON

Download + EDA

Kaggle pull, load transactions, build the household × commodity matrix, first sparsity + coverage check.

TUE

Baseline + freeze eval

RFM segments, popularity + segment-popularity baselines, temporal split w1–91 / 92–96 / 97–102, evaluate() frozen.

WED

Improved method

Implicit ALS (Hu-Koren-Volinsky) on log-spend. Optional LightFM hybrid with commodity + department features.

THU

The money metric

Coupon-redemption value on hold-out as incremental-sales proxy. Category coverage + intra-list diversity as secondary.

FRI

4-min pitch

Problem · Dataset · Research question + method · Evaluation + risks. Rehearse twice before the session.

METRICS — ≥ 2 relevance + 1 beyond-accuracy

Recall@3

Did we surface the right campaigns?

NDCG@3

Were they ranked correctly?

Redeemed-value uplift

The money metric (incremental-sales proxy)

RISKS WE STATE UP FRONT

Small panel — 2,500 households means wide CIs.

Proxy, not causal — redemption ≠ true uplift without a control.

Baseline risk — RFM is strong; the improved method must earn the win.